

SHANMUGARAJ

Full-Stack Developer (MERN) | 2025 ECE Graduate | Software Application Development

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SUMMARY

Full-stack developer and 2025 Electronics & Communication Engineering graduate with hands-on experience building and deploying end-to-end applications using React.js, Node.js, PostgreSQL, and MongoDB. Experienced in designing RESTful APIs, implementing JWT-based authentication and role-based access control (RBAC), and building scalable, production-ready systems. Strong foundation in data structures, databases, operating systems, and computer networks, with 150+ LeetCode problems solved. Passionate about AI-powered applications, large language models (LLMs), and building reliable products that solve real-world problems.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript (learning), C++, SQL, Python, HTML5, CSS3

Frontend: React.js, Redux, Bootstrap, Responsive UI Design

Backend: Node.js, Express.js, REST API Design, MVC Architecture, Mongoose, Prisma ORM

Databases: PostgreSQL, MongoDB, Query Optimization, Relational Schema Design

Security & Testing: JWT, RBAC, bcrypt, Input Validation, Unit Testing, Integration Testing

Cloud & DevOps: AWS, Azure, Git, GitHub, Vercel, Postman, CI/CD (GitHub Actions - learning), Agile/Scrum

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design (learning)

Productivity & Soft Skills: Google Workspace (Docs, Sheets, Slides), self-motivated, organizational skills, independent & team collaboration, adaptability under ambiguous priorities

EDUCATION

B.E., Electronics & Communication Engineering

2021 - 2025 (Graduated)

Jerusalem College of Engineering, Chennai

CGPA: 7.5/10 | No Arrears | Coursework: DSA, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Software Intern | NSIC - Technical Services Center

2024

- Developed and debugged C/C++ applications for microcontroller platforms, building strong foundations in low-level programming, memory management, and systematic root-cause analysis.
- Integrated AI/ML inference into embedded systems and performed structured testing and validation, applying the same debugging discipline to full-stack project development.

PROJECTS

Task Flow - Team Task Manager | React.js, Node.js, Express.js, PostgreSQL, Prisma ORM, JWT, RBAC

- Built and deployed a full-stack task management platform with a Kanban board, task filtering, overdue task tracking, and role-based collaboration for team project management.
- Designed MVC RESTful APIs with JWT authentication and admin/member RBAC; optimized PostgreSQL queries using Prisma ORM to improve performance and scalability.
- Kept the codebase clean and production-scalable using modular React components and Git feature branching.

Vinayaga Construction Management System | React.js, Node.js, MongoDB, Mongoose, RBAC, OOP

- Built a multi-role platform (admin / manager / staff) using OOP principles, applying least-privilege RBAC and bcrypt + JWT credential handling to fix OWASP Broken Access Control issues.
- Wrote unit and integration tests covering all major features prior to Vercel deployment, ensuring reliability in a real-world construction workflow context.

College Transport Tracking System | Flutter, Dart, Google Maps API, REST API, Real-Time Geolocation

- Independently learned Flutter/Dart to build a cross-platform live vehicle-tracking app using the Google Maps API; diagnosed and fixed real-time map lag by optimizing REST API polling frequency.

PROBLEM SOLVING

- Solved 150+ LeetCode problems across Arrays, Strings, Trees, Graphs, and Dynamic Programming, with strong command of time and space complexity analysis applied to backend optimization.
- Studying distributed systems and system design fundamentals: scalability, caching, load balancing, database sharding, microservices.

ACHIEVEMENTS

- 2nd Place, College Web Development Competition - built a fully responsive, functional web app recognized among all competing teams.
- Deployed 2 live full-stack applications on Vercel featuring RBAC, JWT authentication, and both SQL and NoSQL databases.